# AMYPO — Campus Policy & Placement Copilot

Offline-first grounded question answering over college records. One command starts it; no third-party API is used at any point.

## Problem

Students and staff ask answerable questions of policy documents and structured records, but generic assistants are unavailable offline, cost money per call, and answer without evidence. AMYPO answers only from the campus corpus, cites the record behind every claim, abstains when the evidence is missing, and runs inside an 8 GB budget on one machine.

## Approach

Hybrid retrieval (BM25 + dense fusion with a feature reranker) over 6 300+ chunks; a complexity classifier that routes each question to the cheapest adequate path; a grounding verifier that gates every claim on lexical support and numeric fidelity; a scope-keyed semantic cache; and deterministic placement answering where eligibility must be decided, not generated. The local model degrades in three tiers — phi3:mini, a smaller model, then extractive quoting — so a RAM or bandwidth drop changes the route, never the availability.

## Innovations

Self-optimizing router that retrains from thumbs feedback on a fixed holdout; claim-level citation highlighting with unsupported claims flagged; abstention that names the closest records and asks a clarifying question; conflict detection that surfaces a superseded policy instead of silently picking a winner; predictive cache pre-warming at ingest time; a per-query energy and INR line item against an always-cloud baseline; multilingual input detection; PDF eligibility export; and a readable audit trail for every staff cohort query.

## Headline numbers

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|---|---|
| Answer accuracy | **75.2%** |
| Groundedness | **100.0%** |
| Hallucination rate | **8.3%** |
| Abstention (unanswerable) | **39.1%** |
| Retrieval MRR / S@1 | **0.798 / 75.9%** |
| Routing agreement | **100.0%** |
| Cost vs always-cloud | **98.9% saved** |
| Peak RAM | **188 MB of 8192 MB** |
| Resource scenarios | **6/6 passed** |
| Load test p95 | **2632 ms at 20 users** |

From **docs/eval/report.json** (run 2026-10-18T07:54:42Z), reproduced by **make eval**.

Stack: Python 3.12 · FastAPI · MySQL 8 · scikit-learn · numpy · optional FAISS/sentence-transformers · Next.js 14 · Ollama (phi3:mini), all local. Models: phi3:mini 3.8B q4 (~2.2 GB, MIT) with a deterministic extractive fallback; see docs/model_card.md for licences and known limitations.